

UrjaMitra AI

How We Upgrade Your Factory Floor

MSME Idea Hackathon 6.0

A ZERO-DISRUPTION UPGRADE TO INDUSTRY 4.0



STEP 1

Your Factory Floor

We start by adding simple, non-invasive sensors to your existing machinery. No wire cutting, no machine downtime.



Clip-on Power Sensors

Split-core sensors measure current instantly, showing exactly how much power the machine draws.



Machine Vitals

External sensors monitor motor temperature and vibrations to feel the "health" of the machine.

Textile Looms

CNC Machines

Packaging



STEP 2

The Smart Hub

A small, affordable IoT device collects all sensor data and securely transmits it over your factory's Wi-Fi.



Data Cleaning

The hub instantly removes electrical noise and calculates the true power consumption on the spot.



Secure Transmission

Data is sent seamlessly to the cloud using lightweight, secure industrial protocols (MQTT).

ESP32 Chip

Wi-Fi / 4G

MQTT Protocol



STEP 3

The AI Brain

Our cloud engine analyzes the data 24/7, creating a "Digital Twin" that thinks and predicts issues for you.



Finds Wasted Power

The AI detects when a machine is turned on but producing nothing, flagging invisible energy waste.



Predicts Breakdowns

By noticing tiny changes in vibration and heat, it warns you before a motor burns out or a bearing breaks.

AI Rule Engine

Timeseries DB

Prisma ORM



STEP 4

Your Dashboard

You get a clean, easy-to-use app that translates complex data into simple actions and exact rupee savings.



Live ROI & Savings

See exactly how much money you save every day by cutting down idle power wastage.



Maintenance Alerts

Receive SMS/App notifications saying "Service Machine A today to prevent a breakdown tomorrow".



Green Certification

Automatically generate carbon reduction reports required for ZED (Zero Defect, Zero Effect) compliance.

Next.js Web App

Mobile Friendly